

Bag of Words Note-Taking Framework

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Tuesday 23rd September, 2025 - 09:41

1 Summary

This week we explored the Bag of Words approach to creating word embeddings. Bag of Words is a counting based method, this differs from some of the other approaches we've looked at such as SkipGram which was a prediction based method. Before we start with the process of Bag of Words, we first have to pre-process the data. This pre-processing has three steps. First, we tokenize everything in the training data. The next step is to then go in and filter out all of the stopwords. These are words that occur frequently, but don't really carry meaning, such as the, a, it, etc. Finally we can then do stemming or lemmatizing on our remaining tokens, this just refines the tokens further. Now we get into Bag of Words, it starts by counting all of the words in the training data and creating a matrix. This matrix contains each word and how many times it exists in the data (Bag of Words). Once we have this count or vocab we can use this to create embeddings and draw meaning from our words. It can also be used to predict what a text is talking about by identifying frequent used words within the text (Getting started with Natural Language Processing: Bag of words, 3:35). We see this used in Siamese Continuous Bag of Words. In this process we use the Bag of Words process and aim to create embeddings specifically for the purpose of averaging them within sentences. This helps with creating a model that can predict surrounding sentences, which is exactly what Siamese Continuous Bag Of Words aims to do. This method sought to expand and optimize sentence prediction, because existing methods such as Word2Vec are not optimized for it and thus perform worse (Siamese CBOW: Optimizing Word Embeddings for Sentence Representations).

2 Attention Grabbers

1. One thing that caught my attention was towards the end of the first video, he goes into how to code the bag of words model. A lot of the things we've seen so far have been purely concepts, no code so I liked that this showed how you could code the model out. (Getting started with Natural Language Processing: Bag of words)

2. I really like how the second video had a paragraph of words to start, then as he talked about each pre-processing step the paragraph changed to reflect that step. It was really nice visual to kind of see how these things actually effect a text.
3. Something that jumped out to me in the paper was that Siamese CBOW is used to predict sentences rather than words. This is different than all the other models we've looked at so far.
4. I think it's interesting that there is no "one way" to create embeddings. The paper shows too that certain ways of creating/learning embeddings are more optimized for certain tasks than others.
5. I think it's interesting that despite Siamese CBOW having Bag of Words in the name, it is also very reminiscent of SkipGram. It predicts the preceding and following sentence using a center sentence, similar to how SkipGram predicted surrounding words around a center word.

3 Questions

1. In the first video it talks about the model just ignoring words that it doesn't recognize. What if one of the words it doesn't recognize is important, wouldn't this lead to a loss of understanding by the model?
2. The paper talks about the Siamese CBOW learning embeddings by predicting the surrounding sentences. Do you have to tokenize differently when predicting sentences instead of words?
3. I noticed that Siamese CBOW performs best on tweet data. Does this have to do with most tweets being about one topic, so the sentence prediction is easier?
4. Do most of the major LLMs focus more on predicting the next word or predicting sentences when creating their embeddings?
5. What are the main benefits to focusing on sentence prediction as opposed to just predicting one word at a time.